Guidelines for Master Thesis students (NetGroup)

Beginning of the Thesis Work

On-boarding Process:

When starting a thesis, each student will be given access to different platforms.

First, as soon as the thesis is confirmed, the supervisor (or co-supervisor) will send to the student an invitation for the **Slack** workspace of the NetGroup, where a dedicated private channel (e.g., "*tesi-surname-topic*") will be created. This channel <u>must</u> be used as the main communication channel with the supervisor and co-supervisor(s).

Second, the student will be enabled to access **GitLab1**, a private GitLab server of the Politecnico di Torino (for more information on this point refer to doc). Here each student will get a dedicated repository for their activity. Thesis students are free to use this repository as they prefer during their work. However, what they are required to upload on it is:

- the material consulted during their work;
- the documentation produced during their work;
- the PowerPoint presentations shown during the meetings;
- the Latex source files of the final thesis document.

The code is instead commonly committed on a separate repository (unless differently suggested by the co-supervisor). The personal repository, usually named like "*Year-Surname_Name*", is also the place where the supervisor(s) and co-supervisor(s) will share any useful literature and documentation.

Laboratory Access:

For students requiring laboratory access, they must agree this with the supervisor, and get all the information required to send the request electronically with an online procedure.

<u>We remark that spots in the laboratory are limited, and we cannot guarantee a spot for each student</u>. It is the responsibility of students to interact with other researchers and PhD students in the laboratory to understand the availability of desks, which may vary. For this reason, access to the laboratory is recommended especially if there are particular reasons (e.g., interest in PhD, require access to a workstation).

Carrying Out the Thesis Work

Managing Research Milestones:

To keep progressing on the research and to remain aligned on the topic, different **milestones** will be defined in the thesis work. Whenever a milestone is reached, the student, after discussing it with the co-supervisor(s), should request a more structured meeting to present the status and discuss how to proceed.

Effective milestone management is crucial for maintaining research momentum. The student is expected to demonstrate its <u>autonomy</u> during the thesis, trying to solve by themselves the encountered problems. However, prolonged stagnation on a single issue should be avoided. If this happens, the student should seek targeted discussions with the co-supervisor(s) and supervisor through Slack.

Preparation for meetings:

To maximize the value of meetings, the student is expected to prepare for it in advance. Some general guidelines are:

- develop a (PowerPoint) **presentation** with a very brief context (multiple thesis are conducted in parallel, short summary eases the supervisor(s) in framing the discussion), achieved milestone(s), key development and model choices, challenges, and prospective research directions;
- formulate precise discussion points to foster constructive feedback and exchange;
- meetings with supervisor(s) will not be many, make the most out of them!

Material prepared for meetings should be uploaded to GitLab1.

Access to Scientific Literature:

During the thesis, students may need access to scientific literature to perform their work. This access is granted if browsing from the university network. If for any reason this is not an option, scientific literature could be accessed through https://pico.polito.it/, using the institutional credentials.

Conclusion of the Thesis Work

Deadlines and subscription to final exam:

It is responsibility of the student to keep track of the many deadlines involved in the thesis (e.g., request of conclusion, upload of final version, payment of the final exam) and remind them to the supervisor(s), giving him reasonable time to complete possible action under his responsibility.

If some deadlines are missed (e.g., payment of final tax) there is no possibility for the student to be admitted at the final exam!

The conditions that we set for you to request the thesis conclusion officially, to be completed by the conclusion request deadline of the specific final exam call, are the following ones:

- to have completed the minimum objectives of the thesis work;
- to have validated the results obtained and the completion of the objectives with the supervisor;
- to have written approximately 70%-80% of the thesis document.

Thesis Composition and Academic Writing:

Commonly, each thesis requires first a phase of research and modelling/implementation, and only after the phase of writing. However, continuous writing and documentation of each step is very important to arrive prepared to the writing phase.

Some guidelines to write the thesis are:

- the language depends on the path the student is attending (e.g., Cybersecurity, Computer Engineering). If the path is **only** in English (as Cybersecurity is), then the thesis **must** be written in English. If the path is **mixed** English-Italian, then the language of the thesis can be both (English or Italian). However, English (US) is recommended;
- writing in LaTeX (e.g., using Overleaf or locally installed editors/compilers) is recommended even if not mandatory;

- it is important for students to use their own voice while writing. The final version of the thesis must pass a plagiarism (and AI) check in order to be accepted. If these thresholds are not met, the supervisor cannot approve the student's request;
- use a spell checker to verify the correctness of the writing (if English, use the US dictionary and not the UK one);
- pictures are often more effective than words when describing architectures or complex concepts, a useful tool in this sense could be https://draw.io/.

To structure the thesis, students should analyze and follow the same model used by previous colleagues. Their thesis could be searched on Home - Webthesis, filtering based on the supervisor. Here, we provide a generic structure as a starting point:

- Introduction (1 chapter): to briefly introduced the context and addressed problem
- Background (1-3 chapters, commonly 2): to describe the thesis research area and/or used tools (e.g., the area of "network security automation" with VEREFOO as tool would be included in a single background chapter)
- Thesis Objectives (1 chapter): to describe very shortly (e.g., 2-3 pages) the objectives of the thesis work, in view of the illustrated background
- 1-3 chapters related to the specific Thesis Work, whose number and size may vary depending on the specific thesis. They often describe the Approach and the Model
- Implementation and Validation (1 chapter): to describe how the investigated approach has been implemented and validated experimentally
- Conclusions and Future Work (1 chapter): to draw conclusions and discuss possible future directions

Summary:

The summary is to be submitted with the final version of the thesis (one week before the discussion). It should be up to three pages and provide a clear explanation to help the other members of the committee, who may not be specialists in the field, understand. The important elements to convey are (i) the addressed problem, (ii) the proposed solution/methodology, and (iii) the results obtained. It should clearly highlight students' contributions.

There is not a single possibility in terms of template, however students can access some examples from previous students and a recommended template here: examples. Including one or two figures is acceptable if they help clarify aspects of the text (e.g., illustrating a system architecture). However, excessive use of figures should be avoided.

Final presentation:

The final presentation is part of the final exam, and part of the graduation score depends on its quality. For this reason, it should be conducted as professionally as possible. Students should follow these guidelines:

- prepare 10-15 slides (title and conclusion included);
- use a spell checker to avoid typos;
- slides should not be "text-heavy", but they serve as a guide for speaking, use visuals instead of long text whenever possible;
- make sure that presentation fits within the available time slot (15 min for presentation + Q&A, so around 12 min for presentation only);
- students can ask (if desired) for a final meeting for rehearsal of the presentation, where the co-supervisor can provide their feedback;

- the chosen language follows the same principles of thesis writing;
- slides should be formatted for readability, so use large fonts (minimum 22-24pt for text, 30pt for titles, smaller text accepted for diagrams), use high-contrast colors, keep slides concise.

A suggested structure for the presentation is the following one:

  - Title slide: Candidate's name, thesis title, advisor(s)
  - Problem Introduction and Thesis Contribution (1-3 slides): From a general overview to the specific addressed problem and thesis contributions (it is very important to make the committee understand what the exact **contributions** of the work are!)
  - Methodology (a few slides): Explain the work carried out by the candidate
  - Results (a few slides): Present the key findings
  - Conclusion slide: Summarize findings and outline potential future work

Some examples from previous students can be found here: examples.